THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 7th term is: 1

(a) 21 **(b)** 29 **(c)** 13 **(d)** 34

2. What is the smaller angle between the hands of a clock at 3 o'clock? 1

(a) 90 **(b)** 60 **(c)** 120 **(d)** 270

3. Add 14 and its reverse 41. The sum is: 1

(a) 44 **(b)** 77 **(c)** 55 **(d)** 66

4. In a pictograph one symbol stands for 10 units. How many symbols (counting a half symbol as half) are needed to show 45? 1

(a) 4 **(b)** 4.5 **(c)** 5 **(d)** 450

5. Which of the following is divisible by 4? 1

(a) 3518 **(b)** 3516 **(c)** 3517 **(d)** 3519

6. The area of a triangle with base 10 cm and height 8 cm is (in cm^2): 1

(a) 48 **(b)** 50 **(c)** 40 **(d)** 30

7. $\frac{1}{2} + \frac{1}{3} =$ 1

(a) $\frac{2}{5}$ **(b)** $\frac{5}{6}$ **(c)** 1 **(d)** $\frac{1}{6}$

8. A circle has centre O and radius 5 cm. A point P is 9 cm from O. The point P lies: 1

(a) at the centre of the circle **(b)** outside the circle
(c) on the circle **(d)** inside the circle

9. A rhombus (not a square) has how many lines of symmetry? 1

(a) 4 **(b)** 2 **(c)** 1 **(d)** 0

10. -12 + 7 = 1

(a) -5 **(b)** -3 **(c)** -7 **(d)** 9

11. A complete graph has 3 points, each joined to every other point. How many lines are there? 1

(a) 4 **(b)** 9 **(c)** 6 **(d)** 3

12. A ray turns anticlockwise through 20° from the horizontal line to the right. The type of angle formed is: 1

(a) obtuse **(b)** straight **(c)** right **(d)** acute

13. Kaprekar step: for 3752, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 5175 **(b)** 5265 **(c)** 5166 **(d)** 5184

14. A bar graph has scale 1 cm = 5 students. A bar showing 40 students will have height (in cm): 1

(a) 10 **(b)** 8 **(c)** 200 **(d)** 35

15. Which pair is a pair of co-prime numbers? 1

(a) 6 and 9 **(b)** 12 and 18 **(c)** 10 and 25 **(d)** 8 and 15

16. A right-angled triangle has legs 6 cm and 8 cm. Its area (in cm^2) is: 1

(a) 24 **(b)** 18 **(c)** 30 **(d)** 32

17. $\frac{3}{4}$ - $\frac{1}{3}$ = 1

(a) $\frac{5}{12}$ **(b)** 2 **(c)** $1\frac{1}{12}$ **(d)** $\frac{1}{3}$

18. Two points A and B are 8 cm apart. The perpendicular bisector of AB meets AB at a point which is at what distance from A (cm)? 1

(a) 8 **(b)** 3 **(c)** 4 **(d)** 5

19. The smallest angle through which a regular pentagon must be rotated to coincide with itself is (degrees): 1

(a) 72 **(b)** 62 **(c)** 102 **(d)** 82

20. 5 - 12 = 1

(a) -7 **(b)** 17 **(c)** -9 **(d)** -5

21. The sum of the triangular numbers 10 and 15 is: 1

(a) 25 **(b)** 21 **(c)** 29 **(d)** 24

22. Angles $\angle AOB = 35^\circ$ and $\angle BOC = 40^\circ$ are adjacent (they share the arm OB, and lie on opposite sides of it). Then $\angle AOC$ is: 1

(a) 75 **(b)** 105 **(c)** 5 **(d)** 85

23. In the hailstone (Collatz) sequence, the number after 18 is: 1

(a) 10 **(b)** 36 **(c)** 55 **(d)** 9

24. Using the table, how many more are there of Banana than Guava? 1

Item	Number
Guava	20
Banana	40
Grapes	35

(a) 20 **(b)** 15 **(c)** 60 **(d)** 25

25. Which of these is a pair of twin primes? 1

(a) 7 and 11 **(b)** 19 and 23 **(c)** 11 and 13 **(d)** 13 and 17

26. A square has area 64 m^2. Its perimeter (in m) is: 1

(a) 32 **(b)** 40 **(c)** 48 **(d)** 24

27. Which is the greatest of $\frac{3}{5}$, $\frac{5}{8}$, $\frac{7}{12}$, $\frac{2}{3}$? 1

(a) $\frac{7}{12}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{3}{5}$

28. The diagonals of a rectangle are each 10 cm long and meet at O. The length OA (cm) is: 1

(a) 10 **(b)** 6 **(c)** 5 **(d)** 4

29. The mirror image of the letter A in a vertical mirror to its right looks like: 1

(a) a reversed letter **(b)** a different letter
(c) an upside-down letter **(d)** the same letter

30. On the number line, the integer -6 lies 6 units to the left of 0. The integer is: 1

(a) 6 **(b)** -6 **(c)** -7 **(d)** -5

31. What is the sum of the first 15 natural numbers $1 + 2 + 3 + \cdots + 15$? 1

(a) 135 **(b)** 225 **(c)** 120 **(d)** 240

32. You face North and turn clockwise through 90 degrees. What further clockwise turn (in degrees) brings you back to facing North? 1

(a) 90 **(b)** 360 **(c)** 270 **(d)** 180

33. The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 57: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 4

34. The total number of students in the table is: 1

Sport	Students
Hockey	30
Tennis	20
Cricket	25

(a) 55 **(b)** 75 **(c)** 85 **(d)** 70

35. The number of prime numbers between 10 and 30 is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

36. The area of a rectangle is 144 cm^2 and its length is 16 cm. Its breadth (in cm) is: 1

(a) 9 **(b)** 7 **(c)** 25 **(d)** 18

37. $1\frac{1}{2} + 2\frac{1}{3} =$ 1

(a) $3\frac{1}{2}$ **(b)** $3\frac{2}{5}$ **(c)** $3\frac{5}{6}$ **(d)** $3\frac{1}{6}$

38. In rectangle ABCD, diagonals meet at O and $\angle AOB = 60°$. Then $\angle OAB$ is (degrees): 1

(a) 120 **(b)** 60 **(c)** 70 **(d)** 30

39. A figure coincides with itself after turns of 90, 180, 270, ... degrees, up to a full turn of 360°. Its order of rotational symmetry is: 1

(a) 6 **(b)** 5 **(c)** 3 **(d)** 4

40. The number of integers between -5 and 4 is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 7

41. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 16th one is: 1

(a) 32 **(b)** 256 **(c)** 240 **(d)** 272

42. 4 rays start from a point O, no two of them in the same direction. If every pair of rays makes one angle, how many angles are there? 1

(a) 12 **(b)** 3 **(c)** 6 **(d)** 4

43. The digit sum of 4567 is: 1

(a) 29 **(b)** 22 **(c)** 18 **(d)** 31

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44. The number which occurs most often in the data 5, 2, 3, 2, 4, 2, 3, 2, 4, 1, 1, 3 is: 1

(a) 3 **(b)** 1 **(c)** 2 **(d)** 4

45. The prime factorisation of 72 has how many prime factors (counting repeats)? 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 6

46. Fencing a rectangular plot of perimeter 80 m costs ₹15 per metre. The total cost is (₹): 1

(a) 1200 **(b)** 1215 **(c)** 1185 **(d)** 1300

47. Arrange in ascending order: $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$. 1

(a) $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$ **(b)** $\frac{3}{4}, \frac{5}{6}, \frac{2}{3}$

(c) $\frac{2}{3}, \frac{5}{6}, \frac{3}{4}$ **(d)** $\frac{3}{4}, \frac{2}{3}, \frac{5}{6}$

48. A circle of radius 3 cm and a concentric circle of radius 7 cm are drawn. The width of the ring between them (cm): 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 6

49. A regular polygon has 7 lines of symmetry. The number of its sides is: 1

(a) 6 **(b)** 7 **(c)** 9 **(d)** 8

50. A diver at -15 m (below sea level) swims up 6 m. His new position (m) is: 1

(a) -11 **(b)** 3 **(c)** -9 **(d)** -7

END OF QUESTION PAPER **Total: 50 Marks**

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